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Ysbyty Gwynedd Anaesthetics Department Paediatric Patient Controlled Analgesia (PCA) and Nurse Controlled Analgesia

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1. INTRODUCTION

Paediatrics Patient Controlled Analgesia (PCA)

Patient-controlled analgesia (PCA) is a commonly used modality for acute pain management in the paediatric population (Nelson et al., 2010). It allows the patients to self-administer a pre-programmed intravenous bolus dose of opioid using a computerized infusion pump. The device is set with a minimal time interval, lockout and dosage bolus. A background continuous infusion dose of opioid can also be programmed. However, in most cases there is no background rate. The benefits of PCA use include improved pain scores and patient satisfaction compared to non-patient-controlled forms of opioid administration (American Society of Anaesthesiologists Task Force on Acute Pain Management 2012, McNicol et al., 2010, Cravero et al.2019).

PCA use is generally accepted as safe in the paediatric population (McNicol et al., 2015, Faerber et al., 2017, McDonald & Cooper, 2001). It can be offered to any child who is able to grasp the concept of pressing the button to help relieve the pain (Verghese & Hannallah, 2010). PCA can be used for children 6 years and older (Nelson et al., 2010). Although opioid consumption has been found to be higher with PCA compared with non-patient-controlled regimes, generally severe adverse effects have not been found to be higher with PCA (McNicol et al., 2015, Faerber et al., 2017). PCA use has been associated with higher rates of pruritus (McNicol et al., 2015) compared to conventional opioid administration.

Successful use of the PCA as a method of patient-centred pain management requires adequate education of patients, families and nursing staff. The concept of pushing the button by the patient specifically for pain, requires reassurance regarding safety measures, and teaching around potential opioid-related side effects prior to the initiation of the PCA (Gay et al, 2020).

Nurse Controlled Analgesia (NCA)

Many principles of NCA and PCA are similar, that is, a button is pressed to deliver a dose. However, in the NCA mode the nurse caring for the patient may press the button to give a bolus dose on the basis of a request of analgesia, pain severity scoring or in anticipation of pain (e.g., physiotherapy). To ensure safety there are important differences namely a **longer lockout period and a background infusion** in the prescription. The nursing staff/ anaesthetist should decide if an NCA or PCA best suits a child. Usually, children under the age of 6-years are managed with an NCA, 6-years and above the child can often understand the concept of PCA (Gay et al, 2020).

Some general principles which apply to children

- Paediatric analgesia needs to be calculated on their weight (in kilograms), rounding off to make volume calculations easy is helpful.

- Children do not like intramuscular (IM) injections, and should only be used in extreme circumstances. As well as being disliked by children, the effects are often unpredictable and children may deny having pain to avoid injections.

- Pain is better prevented than treated. Analgesic requirements are lower if children are allowed to wake up comfortably post-operatively or are pre-treated before painful procedures.

- Severe pain is best treated with a continuous method of analgesic administration (e.g., PCA or NCA*).* Mild pain can usually be treated with intermittent bolus doses.

- The parents and the child should be taught the principles of PCA pre-operatively. Parents must be advised of the dangers of pressing the button for the child.

- Only nurses with additional training provided by the inpatient 'Acute Pain Service' can look after children prescribed PCA/NCA.

- PCA/NCA should be used in conjunction with multimodal analgesia e.g., paracetamol.

2. POLICY STATEMENT

BCUHB is committed to ensuring acute pain is managed appropriately in children who are inpatients using PCA and NCA, ensuring the opioids are prescribed administered and monitored appropriately.

3. AIMS

The aim of this guidance is to assist Anaesthetist and healthcare professionals within the department to promote effective of post-operative pain in children using PCA and NCA. This guideline refers to PCA/NCA for the relief of acute pain only.

4. OBJECTIVES

To facilitate the safe and effective use of paediatric PCA/NCA and the optimal management of complications and side effects.

5. SCOPE

Ysbyty Gwynedd clinical staff involved in the care of children with PCA and NCA, who undergo additional training provided by the Acute Pain Service.

6. ROLES AND RESPONSIBILITIES

- Patient selection and prescribing is the responsibility of the Anaesthetist.
- Patient education is the responsibility of the anaesthetist and paediatric nurses
- Setting up the device is the responsibility of the anaesthetist, recovery staff and the acute pain team.
- Ward team are responsible for monitoring the child
- Pharmacy team is responsible for supply of medication
- Medical electronic are responsible for the maintenance and servicing of the devices
- BCUHB is responsible for adequate staffing and safety
- Acute pain team is responsible for device training and education.
- Procurement and ward staff for the supply of disposable administration sets
- Acute pain team should be available within normal working hours and respond to requests for advice and assessments and patient reviews.
- Out of hours contact the 'Anaesthetist on Call' (Bleep 063)

7. MAIN BODY

Patient Selection

A PCA is suitable for a child who requires opioid analgesia for a 24-hour period or are expected to have severe acute pain and for whom the oral route is not appropriate. Prior to setting up the infusion the anaesthetist or acute pain team will consider the suitability of PCA/NCA for each individual patient.

Indications for PCA

If a child prescribed a PCA must:

- Understand the concept of PCA, that is, pressing the button when it hurts.
- Physically be able to press the button.
- **Otherwise, children under 6 years or those unable to understand the concept of PCA, should use an NCA (Nelson et al., 2010).**

Indications for NCA

Treatment of acute pain in children, greater than 1 year old or when the child is not capable of using a PCA.

Contraindications

Absolute contraindications to PCA/NCA are:

- Known opioid allergy (very rare)
- Lack of safe environment for monitoring

Extra caution is advised when using PCA/NCA in children with certain medical conditions:

- Raised intra-cranial pressure (ICP): known of suspected raised ICP of any cause represents a relative contraindication to PCA/NCA analgesia.

- Severe respiratory disease: analgesia should be used with caution in these children, although PCA delivery may confer decreased risk compared to other routes of delivery.

PCA dosing

The PCA pump is programmed with a bolus dose, lockout time, background infusion and 2-hour limit. The 2-hour limit when starting a PCA is generally calculated as 80% of the maximum allowable 2-hour dose. The majority of PCAs are administered without a background infusion. There is currently no strong evidence to using one opioid over another for PCA purposes (Cravero et al., 2019).

Prescription

A PCA/NCA should be hand written in the PRN section on a Paediatric Drug Chart, with ondansetron and naloxone based of the weight of the child. An adult PCA sticker should **NOT** be used.

The anaesthetist prescribing the PCA/NCA should check the child has no other concurrent opioids prescribed to minimize the risk of accidental administration and adverse side effects.

Morphine is considered the “gold standard” for intravenous analgesia and is therefore usually the opioid of choice. Fentanyl is not recommended as an initial opioid of choice given its increased potency and shorter half-life (Gai et al., 2020).

No supplementary opioids (e.g., codeine/tramadol) should be administered which the patient is receiving PCA/NCA, unless specifically ordered by an Anaesthetist or Acute Pain Service. The US Food and Drug Administration (FDA) issued an updated warning about codeine and tramadol use in children in 2017, because of reports of ultra-rapid metabolisers and severe respiratory depression.

Changes to the prescription

Standard drug concentrations and standard pump programs should be used at all times, unless approved by a Consultant Anaesthetist in exceptional circumstances.

Standard Drug Calculations

Morphine

	Morphine	
PCA Mode	PCA	NCA
Final conc.	20mcg/kg/ml	20mcg/kg/ml
Loading Dose	Zero	Zero
Bolus Dose	1ml	1ml
Lockout time	5 minutes	15 minutes
Background rate	0 ml/hr.	1ml/hr.

Calculating the Morphine PCA/NCA Concentration

The dedicated paediatric morphine program is 1 mg/kg i.e., the child's weight in kg should equal milligrams of morphine.

E.g., child's weight = 30kg, morphine =30mg (to a max of 50mg) made up to 50ml with 0.9% Sodium Chloride

Final concentration = 20mcg/kg/ml

Child Weight in Kilograms (Kg)	Morphine (1mg/ml) Use 50mg/50ml ampoules	0.9% NaCl Volume (ml)
10	10 ml	40 ml
20	20 ml	30 ml
30	30 ml	20 ml
40	40 ml	10 ml
≥50	50 ml	0 ml

Fentanyl

	Fentanyl	
PCA Mode	PCA	NCA
Final conc.	**0.2mcg/kg/ml**	**0.2mcg/kg/ml**
Loading Dose	Zero	Zero
Bolus Dose	1ml	1ml
Lockout time	5 minutes	15 minutes
Background rate	0 ml/hr.	1ml/hr.

Child Weight in Kilograms (Kg)	Fentanyl (50mcg/ml) Volume (ml) Use 500mcg/10ml ampoules	0.9% NaCl Volume (ml)
10	2 ml	48ml
15	3ml	47ml
20	4ml	46ml
25	5ml	45ml
30	6ml	44ml
35	7ml	43ml
40	8ml	42ml
45	9ml	41ml
≥50	10ml	40ml

Calculating Fentanyl PCA/NCA Concentration

Child Weight in Kilograms (Kg)	Bolus Dose
10	2 mcg/ml
15	3 mcg/ml
20	4 mcg/ml
25	5 mcg/ml
30	6 mcg/ml
35	7 mcg/ml
40	8 mcg/ml
45	9 mcg/ml
≥50	10 mcg/ml

Adjuvant drugs – the following drugs should be prescribed when PCA/NCA is commenced

Drug	Route	Dose	Frequency	Max Dose
Naloxone	IV bolus	2-4 microgram/kg	Single dose The half-life of naloxone is shorter than opioids therefore a second dose maybe required	200 micrograms
Child Weight in Kilograms (Kg)	**Naloxone Dose**		**Make (400mcg/2ml) ampoule of naloxone to 8mls with 0.9% NaCl = concentration 50mcg/ml**	
10	20-40mcg		0.1 – 0.2 ml	
20	40-80mcg		0.2 – 0.4 ml	
30	60-120mcg		0.3 – 0.6 ml	
40	80-160mcg		0.4 – 0.8 ml	
50	100-200mcg		1– 2 ml	
Ondansetron	Oral/IV bolus	0.1 - 0.15mg/kg	8 hourly	6 mg
Laxatives	For prevention/treatment of constipation. See BNFC for doses			

Other drugs

Regular supplementary non-opioid analgesia (e.g., paracetamol) should be prescribed alongside the PCA/NCA, unless contraindicated. Intravenous administration of paracetamol should be considered for all children where enteral administration is not possible. Patients on a PCA/NCA should have an anti-emetic (e.g., ondansetron) prescribed as required. Anti-emetics may be combined.

Equipment

- An Arcomed PCA (available from recovery), using the dedicated paediatric programme.
- 50ml Luerlock syringe
- A continuous IV infusion of fluid to flush the line – minimum rate 5mls/hr.
- The PCA/NCA should be connected to a dedicated intravenous cannula
- If a loading dose is required it should be given before starting the NCA
- The PCA/NCA set must contain an anti-siphon valve.
- Additional IV fluids through the same line must be connected via an anti-reflux/non-return valve to prevent analgesics refluxing back up the second IV line and subsequently being administered as a large bolus
- Administration sets should be changed as per BCUHB policy
- Each drug refill should be signed for as per BCUHB policy
- The key for the PCA/NCA pump should **not** be left in the device once it is running. Keys should be kept with the controlled drugs keys of the ward.
- When the pump is no longer needed it should be cleaned in line with the decontamination policy sent back immediately to the recovery unit.

In the event of problems relating to the PCA/NCA device, contact the Acute Pain Team (Bleep 115) or out of hours the recovery staff on ext. 841831.

Setting up a PCA/NCA

The majority of PCA/NCA pumps will be prescribed and set up in theatre and recovery.
The PCA/NCA is set up by the anaesthetist, recovery or Acute Pain Team (Monday-Friday 9-5 Bleep 115). Out of hours the on-call anaesthetist is available on Bleep 063. The anaesthetist will have responsibility for assessing the suitability of using the PCA.

Qualified staff who have received training are responsible for renewing the syringe in the PCA. Nursing staff have responsibility for monitoring and assessing the child with a PCA/NCA.

All pumps should be kept locked whilst in use using a dedicated line. The antisiphon line should be connected directly to intravenous access.

An anaesthetist or competent nurse will discuss the use of PCA/NCA with the child and family prior to surgery/setting up the infusion.

The child and/or family should be prepared for the PCA/NCA.

- Nursing staff should explain the technique to the child and family
- Parents should be advised in a sensitive manner that they are not permitted to administer a bolus by pressing the button in either technique
- NCA may be changed to PCA or vice versa at any time is appropriate. If the family has any questions that ward staff are unable to answer they should contact the Acute Pain Service on Bleep 115.

Nursing staff should ensure that an age-appropriate pain assessment tool and documentation is available.

- Self-reporting is the gold standard and should be used whenever possible
- Explain the assessment tool to be used (e.g., Wong Baker Faces) to the child.
- The child can practice by scoring any previous painful experiences to determine whether they are able to use the tool
- For infants, patients who are cognitively impaired and those unable to communicate use an appropriate observational tool and involve carers as much as possible

All children receiving intravenous opioids must be supervised by a registered nurse, doctor or appropriately trained ODP.

Any child leaving a clinical area whilst receiving PCA/NCA must be accompanied by a registered nurse, doctor or appropriately trained ODP.

When a PCA/NCA is set up and commenced, a verbal report should be made to the nurse caring for the patient. This should include:

- Details of intra-operative analgesia (if applicable), loading dose and other drugs given.
- Details of drug concentration and pump programme

Additionally, subsequent patient handover, both staff should check that:

- The PCA/NCA prescription chart has been completed correctly.
- The drug being administered corresponds to the treatment chart.
- The pump program corresponds to the most recent entry in the prescription chart.
- Pump settings are within protocol for the age and weight of the child (unless clearly indicated as out of protocol with rationale).
- The patient's pain is being effectively managed.
- The patient is not excessively sedated.
- Drugs to prevent side effects are prescribed e.g., anti-emetics.
- Ensure appropriate doses of naloxone are prescribed.
- Other analgesics such as paracetamol and NSAIDS) are prescribed and administered where appropriate.

Supplementary analgesia e.g., paracetamol and either diclofenac or ibuprofen (where appropriate) should be given regularly if either oral or rectal route is available. Intravenous administration of paracetamol should be considered for all patients where enteral administration is not possible.

No supplementary opioids (e.g., codeine or tramadol) should be administered whilst the patient is receiving PCA/NCA, unless specifically ordered by a consultant anaesthetist.

Patient monitoring

All patients with a PCA/NCA should be cared for by nurses competent in their management.

The following observations should be recorded:

1. Before the child uses the PCA/NCA
2. HOURLY while patients are receiving PCA/NCA
3. For at least 4 hours after discontinuing a PCA/NCA.

Post-operative observations are documented:

- Every 15 minutes for the 1st hour
- Every 30 minutes for the next 2 hours
- Hourly for 4 hours
- Then every 4 hours

Observations should include:

- Respiratory rate
- Oxygen saturation
- Sedation score,
- Pain score (self-reporting should be used whenever possible)
- B/P
- Heart rate
- Temperature.

The following patients should be monitored continuously using pulse oximetry:

- Patients with known sensitivity to opioids
- Patients with respiratory complications or airway difficulties
- Patients whose sedation is repeatedly equal to 2 or more.

Hourly pump readings should also be recorded on the PCA/NCA observation chart. This should include:

- Number of demands
- Number of successful tries
- Amount infused (hourly and running totals)
- The pump should never be re-zeroed whilst in use

If during the night the child is asleep and all the other observations are within normal limits, it is acceptable to omit the sedation and pain assessment and record the fact that the child is asleep.

The amount of drug used or left in the device should be recorded on the observation chart.
The IV site should be checked for swelling, pain and leakage.
PCA usage should be documented on specific PCA documentation (Appendix 1).
Unused Morphine or Fentanyl must be discarded in accordance with BCUHB Medicine Management.
All patients using PCA should be reviewed at least once a day by a member of the Acute Pain Service.

If the patient is in pain:

- Explore the possible causes of pain with the patient
- Involve parents for distraction and reassurance
- Administer prescribed simple analgesics, is appropriate
- Administer prescribed adjunct as appropriate (e.g., anti-spasmodic drugs).
- For PCA: encourage the patient to give a bolus and evaluate the effect after 10 minutes
- For NCA: administer a bolus and evaluate after 10 minutes
- If the pain score remains the same/or greater for more than an hour contact the Acute Pain Team

- The surgical team or the team physician should be contacted if the pain could be indicative of surgical complications, or another cause.

Side effects

Some of the frequently occurring opioid side effects are summarized in Table 1. A common approach to almost all opioid-related side effects is to maximize opioid saving sparing techniques (non-opioid analgesia, regional blocks, and non-pharmacological modalities) and consider opioid rotation. Common side effects include nausea, sedation, respiratory depression and pruritus' (itching). These are side-effects of analgesia rather than of the PCA/NCA

Side effects	Interventions
Sedation	The sedation level of the child must be observed and recorded on the PCA/NCA observation chart at least hourly whilst PCA/NCA is in progress and for 4 hours following discontinuation. The sedation level is scored as follows (see 'PCA Observation Chart': • 0 = Alert • 1 = Mild, easy to rouse • 2 = Easy to rouse, often drowsy • 3 = Somnolent, difficult to rouse If the patient has a sedation score of 3: • STOP the infusion • Ensure the child is well supervised • Continuously monitor saturations using pulse oximetry • Give oxygen by mask until the sedation score improves • Inform the admitting paediatric team or on-call paediatric registrar • Monitor respiratory rate every 15 minutes until improvement in sedation score • Consider other possible causes of sedation e.g., hypoglycaemia • Consider naloxone 2-4 micrograms/kg intravenously • Contact the Acute Pain Team (bleep 115) or the on-call Anaesthetist (bleep 063) to review or change the pump program if required • Document finding and action taken
Respiratory Depression	The respiratory rate of the child must be observed and recorded on the PCA/NCA observation chart at least hourly whilst PCA/NCA is in progress and for 4 hours after it has been discontinued.

	Respiratory depression is defined as: >5 years – Respiratory rate <10 breaths/minute 1 to 5 years – Respiratory rate <15 breaths/minute <1 year – respiratory rate <20 breaths/minutes The definition above is for guidance only and the nurse should take into account: • Depth of respiration • Respiratory effort • Level of sedation If respiratory depression occurs: • STOP the infusion • Ensure the child is well supervised • Continuously monitor saturations using pulse oximetry • Give oxygen by mask until the sedation score improves • Inform the admitting paediatric team or on-call paediatric registrar • Monitor respiratory rate every 15 minutes until improvement in sedation score • Consider other possible causes of sedation e.g., hypoglycaemia • Consider naloxone 2-4micrograms/kg intravenously • Contact the Acute Pain Team (bleep 115) or the on-call anaesthetist to review or change pump program if required • Document findings and action taken • If in respiratory arrest call arrest team (Bleep 2222) and begin resuscitation Naloxone has a half-life of 5-10 minutes; most opioids will have a much longer half-life. Continue to re-assess and give further doses in required.
Nausea, vomiting	The patient should be assessed at least hourly while on PCA/NCA The scores are: • 0 = None • 1 = Mild nausea

	• 2 = Nausea and retching • 3 = Vomiting If the patient complains of nausea or has been vomiting: • Administer an anti-emetic e.g., Ondansetron may be prescribed at 0.1mg/kg (max 4mg) IV 8 hourly PRN. • If in-situ, consider aspirating a nasogastric or gastrostomy tube • Consider stopping oral intake • Consider using another anti-emetic if nausea/vomiting has not improved after an hour e.g., alternatively dexamethasone, metoclopramide may be prescribed. • Consider non-opioid related nausea and vomiting such as sepsis, DKA etc.
Pruritus	The patient should be observed for pruritus (itching) at last 4 hourly, if itching is a problem: • Administer anti-histamine (e.g., chlorphenamine) • Consider naloxone 0.5 microgram/kg intravenously • If pruritus persists, contact Acute Pain Team (bleep 115) and consider converting to fentanyl
Constipation	• Decrease opioid dose is pain is well managed • Promote physical activity (standing, walking) as tolerated by patient • Stool softener and laxatives
Urinary Retention	The patient should be observed for urinary retention. This is defined as the inability to empty the bladder voluntarily for greater than 12 hours with a volume of urine greater than expected age ([age in years + 2] x 30ml) or a palpably distended bladder • Commenced fluid balance chart • Perform bladder scan • Inform medical team • Consider intravenous naloxone 0.5 micrograms/kg • Consider intermittent catheterization

PCA Troubleshooting

Problem	Causes	Intervention
Patient not pressing the button despite pain	Fear related to potential overdose or addiction	Discuss with patient and family the reasons for not using PCA and providing reassurance and education
Patient over-pressing the button	Misunderstanding of appropriate indication for pressing button Press whenever they see a solid light indicating lockout has finished	Reeducation regarding PCA use May not be an appropriate candidate for PCA. Consider another method.
Inadequate pain relief despite appropriate use	The bolus or background dose may be too low, or the interval lockout too low. Specific opioid may not be effective for the patient Assess for non – nociceptive pain (neuropathic, muscle spasm)	Assess whether the bolus doses are effective and if the effect last the 5 minutes lockout. Ensure non-opioid adjuncts are available Consider opioid rotation Institute adjuncts for neuropathic pain e.g., Gabapentin) or muscle spasm (e.g., diazepam) or other non-opioid adjuncts (e.g., ketamine)
Over-sedation	Relative overdose of opioid	Decrease bolus or background by 10-20 %

Technical Issues

If the PCA pump alarms:

- Stop the infusion
- Check the display panel and identify the cause
- If the line is occluded:
    - Check the cannula site
    - Check for kinks or closed camps
    - Flush the cannula
    - Restart the infusion

- If the syringe is empty a new syringe (as per prescription) should be drawn and started by a nurse who is competent to do so.
- Is the alarm due to a flat battery, if so, plug the pump into the main supply?
- If the pump is faulty, it will need to be replaced and returned to Electrical Biomedical Engineering (EBME) with a label describing the fault. Contact Acute Pain Team (bleep 115)
- Any pump which is dropped or damaged must be taken out of service until checked by EBME.

Discontinuing PCA/NCA

The PCA/NCA may be discontinued by a competent member of staff.

The criteria for stopping PCA/NCA include ALL of the following:

- The patient is able to take analgesia via an alternative route and this is prescribed.
- The patient is requiring very few boluses.

The PCA must be stopped by the nurse in the following situation

- Excessive sedation (sedation score of 3)
- Respiratory depression or respiratory arrest
- Pump malfunction

Maintain IV access and observe and record observations for at least 4 hours after discontinuing PCA/NCA.

Any patients with inadequate analgesia or suspected complications should be referred to the Acute Pain Team (bleep 115) or an-call Anaesthetist (bleep 063).

All children with PCA/NCA should be reviewed every day by the Acute Pain Team (except weekends).

8. EQUALITY AND WELSH LANGUAGE

The equality impact assessment (EQIA) did not identify an adverse impact for any protected group. This best practice guidelines are intended as a prescribing resource; therefore, it is not anticipated to impact the use of the Welsh language.

9. WELL BEING FOR FUTURE GENERATIONS

The use of the guideline, for the indication stated in section 1, has no effect on the wellbeing of future generations.

10. ENVIROMENTAL IMPACT ASSESSMENT

An ‘Environment Impact Assessment’ has not been undertaken it is not anticipated to have an environmental impact.

11. RESOURCES

Continuing funding is required for the supply of disposable equipment and the servicing and repair of electronic devices.

12. TRAINING

Time and staffing to allow the training and education of all staff involved in the management of children requiring PCA/NCA. Training records stored locally by BCUHB.

13. IMPLEMENTATION

Successful implementation will require dissemination of information to all applicable staff through the anesthetic department intranet.

14. FURTHER INFORMATION – CLINICAL DOCUMENTATION

See Appendix 1.

15. AUDIT

Anaesthetists and Acute Pain Team will ensure guidance compliance. No audit tool available.

16. REVIEW

The policy will be reviewed and updated every three years, or sooner if required.

17. REFERENCES

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U.S Food and Drug Administration (FDA) (2017) FDA restricts use of prescription codeine pain and cough medicines and tramadol pain medicines in children; recommends against the use in breastfeeding women. Drug and safety communications.

18. APPRENDICES

Appendix 1



GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

Addressograph Label

Patient Controlled Analgesia (PCA) Observation Chart

DOCUMENTATION SHOULD BE COMPLETED HOURLY & FOR 4 HOURS AFTER DISCONTINUING

	Fentanyl PCA	**Morphine PCA**
Date set up		
Pump Number		
Drug Concentration	0.2mcg/kg/ml	20mcg/kg/ml
Protocol (please circle)	Adult >50kg Adult <50kg Paediatrics >50kg Paediatrics <50kg Paediatrics NCA	Adult >50kg Adult <50kg Paediatrics >50kg Paediatrics <50kg Paediatrics NCA
PCA dose	10mcg/ml (>50kg)	1mg/ml (>50kg)
Dose duration/Lockout period	5 mins	5 mins

Date	**Time**	**Patient bolus (ml)**	**Refused bolus (ml)**	**Volume left in syringe (ml)**	**Pain Score (1-10)**	**Nausea Score (1-3)**	**Signature**

Appendix 2

ARCOMED PCA PROTOCOLS

MORPHINE					
PCA Mode	Adult PCA	Adult PCA	Paeds. PCA	Paeds. PCA	Paeds. NCA
Weight	< 50kg	+ 50kg	<50kg	50kg +	50kg/+50kg
Final conc.	20mcg/kg/ml	20mcg/kg/ml	20mcg/kg/ml	20mcg/kg/ml	20mcg/kg/ml
Lockout time	5 minutes	5 minutes	5 minutes	5minutes	15 minutes
Max volume	/	/	18ml/2hrs	20ml/2hrs	5ml/1hr
Background	0 ml/hr.	0 ml/hr.	0ml/hr.	0ml/hr.	1 ml/hr.

FENTANYL					
PCA Mode	Adult PCA	Adult PCA	Paeds PCA	Paeds PCA	Paeds. NCA
Weight	< 50kg	+ 50kg	<50kg	50kg +	50kg/+50kg
Final conc.	0.2mcg/kg/ml	0.2mcg/kg/ml	0.2mcg/kg/ml	0.2mcg/kg/ml	0.2mcg/kg/ml
Lockout time	5 minutes	5 minutes	5 minutes	5minutes	15 minutes
Max volume	/	/	18ml/2hrs	20ml/2hrs	5ml/1hr
Background	0 ml/hr.	0 ml/hr.	0ml/hr.	0ml/hr.	1 ml/hr.

Appendix 3

CALCULATING MORPHINE PCA CONCENTRATIONS

	MORPHINE		
Weight (kg)	**Morphine (1mg/ml) 50mg/50ml**	**0.9% NaCl (ml)**	**Morphine (mg/ml)**
10	10ml	40ml	0.20
11	11ml	39ml	0.22
12	12ml	38ml	0.24
13	13ml	37ml	0.26
14	14ml	36ml	0.28
15	15ml	35ml	0.30
16	16ml	34ml	0.32
17	17ml	33ml	0.34
18	18ml	32ml	0.36
19	19ml	31ml	0.38
20	20ml	30ml	0.40
21	21ml	29ml	0.42
22	22ml	28ml	0.44
23	23ml	27ml	0.46
24	24ml	26ml	0.48
25	25ml	25ml	0.50
26	26ml	24ml	0.52
27	27ml	23ml	0.54
28	28ml	22ml	0.56
29	29ml	21ml	0.58
30	30ml	20ml	0.60
31	31ml	19ml	0.62
32	32ml	18ml	0.64
33	33ml	17ml	0.66
34	34ml	16ml	0.68
35	35ml	15ml	0.70
36	36ml	14ml	0.72
37	37ml	13ml	0.74
38	38ml	12ml	0.76
39	39ml	11ml	0.78
40	40ml	10ml	0.80
41	41ml	9ml	0.82
42	42ml	8ml	0.84
43	43ml	7ml	0.86
44	44ml	6ml	0.88
45	45ml	5ml	0.90
46	46ml	4ml	0.92
47	47ml	3ml	0.94
48	48ml	2ml	0.96
49	49ml	1ml	0.98
≥50	≥50ml	0ml	1ml

Appendix 4

CALCULATING FENTANYL PCA CONCENTRATIONS

	FENTANYL		
Weight (kg)	**Fentanyl (50mcg/ml) 500mcg/10ml**	**0.9% NaCl (ml)**	**Fentanyl (mcg/ml)**
10	2.0ml	48ml	2.0
11	2.2ml	47.8ml	2.2
12	2.4ml	47.6ml	2.4
13	2.6ml	47.4ml	2.6
14	2.8ml	47.2ml	2.8
15	3ml	47ml	3.0
16	3.2ml	46.8ml	3.2
17	3.4ml	46.6ml	3.4
18	3.6ml	46.4ml	3.6
19	3.8ml	46.2ml	3.8
20	4ml	46ml	4.0
21	4.2ml	45.8ml	4.2
22	4.4ml	45.6ml	4.4
23	4.6ml	45.4ml	4.6
24	4.8ml	45.2ml	4.8
25	5ml	45ml	5.0
26	5.2ml	44.8ml	5.2
27	5.4ml	44.6ml	5.4
28	5.6ml	44.4ml	5.6
29	5.8ml	44.2ml	5.8
30	6ml	44ml	6.0
31	6.2ml	43.8ml	6.2
32	6.4ml	43.6ml	6.4
33	6.6ml	43.4ml	6.6
34	6.8ml	43.2ml	6.8
35	7ml	43ml	7.0
36	7.2ml	42.8ml	7.2
37	7.4ml	42.6ml	7.4
38	7.6ml	42.4ml	7.6
39	7.8ml	42.2ml	7.8
40	8ml	42ml	8.0
41	8.2ml	41.8ml	8.2
42	8.4ml	41.6ml	8.4
43	8.6ml	41.4ml	8.6
44	8.8ml	41.2ml	8.8
45	9ml	41ml	9.0
46	9.2ml	40.8ml	9.2
47	9.4ml	40.6ml	9.4
48	9.6ml	40.4ml	9.6
49	9.8ml	40.2ml	9.8
≥50	10ml	40ml	10.0